

How a WhatsApp Message Travels Through the OSI Model

1. Application Layer (Layer 7)

When you open WhatsApp, type a message, and tap **Send**, the Application layer starts the communication. WhatsApp creates the message, adds information such as the sender and receiver details, and requests the network to deliver it.

Example: You type "**Hello!**" and press **Send**.

2. Presentation Layer (Layer 6)

The Presentation layer prepares the data before it is sent. It converts the message into a standard format, **encrypts** it using WhatsApp's end-to-end encryption, and may compress the data to reduce its size. This ensures that only the intended receiver can read the message.

Example: Your "Hello!" message is encrypted into unreadable data while traveling across the internet.

3. Session Layer (Layer 5)

The Session layer establishes and manages the communication session between your phone and WhatsApp's servers. It keeps the connection active while the message is being transmitted and closes the session when communication is complete.

Example: The connection stays active while your message is being sent and delivered.

4. Transport Layer (Layer 4)

The Transport layer breaks the message into smaller pieces called **segments**. It ensures reliable delivery by checking for errors, controlling the flow of data, and reassembling the message correctly at the receiver's device.

Example: If one segment is lost during transmission, it is sent again so the complete message arrives correctly.

5. Network Layer (Layer 3)

The Network layer adds the sender's and receiver's **IP addresses** to the data. It chooses the best path for the message to travel across multiple routers on the internet until it reaches the destination network.

Example: Routers forward your WhatsApp message from your home network to your friend's network.

6. Data Link Layer (Layer 2)

The Data Link layer places the data into **frames** and adds the **MAC addresses** of devices on the local network. It checks for transmission errors before sending the frame to the next device, such as a Wi-Fi router.

Example: Your phone sends the frame to your Wi-Fi router using its MAC address.

7. Physical Layer (Layer 1)

The Physical layer converts the data into **electrical signals, radio waves (Wi-Fi), or light signals (fiber-optic cables)**. These signals travel through the physical network until they reach the destination device.

Example: Your message travels as Wi-Fi signals from your phone to the router and then through the internet.

What Happens at the Receiver's Side?

Once the message reaches your friend's phone, the process happens in **reverse order**:

- **Physical Layer:** Receives the signals.
- **Data Link Layer:** Converts frames back into data.
- **Network Layer:** Confirms the destination IP address.
- **Transport Layer:** Reassembles all message segments in the correct order.
- **Session Layer:** Maintains the communication session.
- **Presentation Layer:** Decrypts the encrypted message.
- **Application Layer:** WhatsApp displays the original message ("Hello!") on your friend's screen.